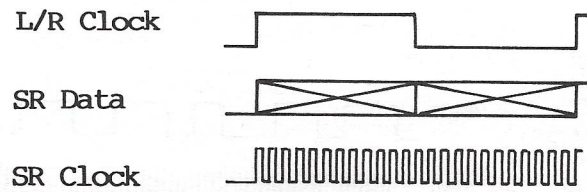
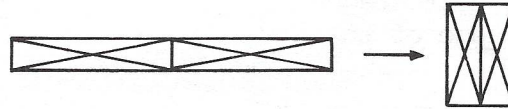


1. Input Signal



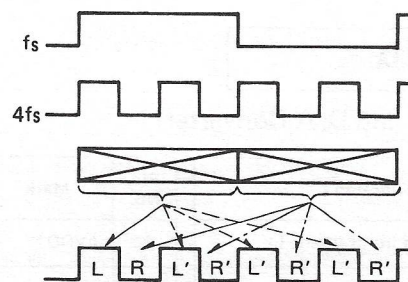
2. S/P Conversion

Converts serial L/R data into parallel L/R data



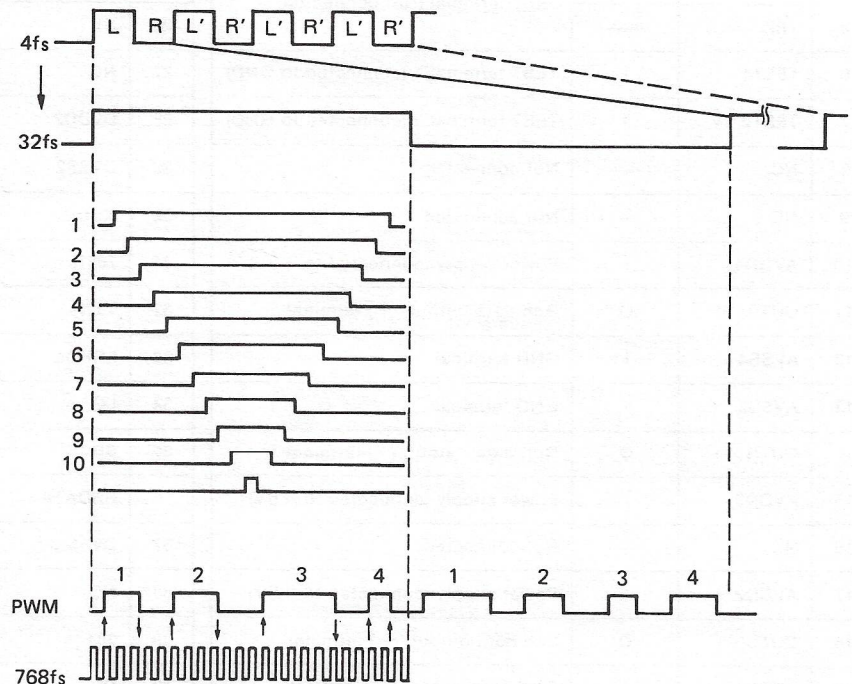
3. D/F Processing

4-hold oversampling



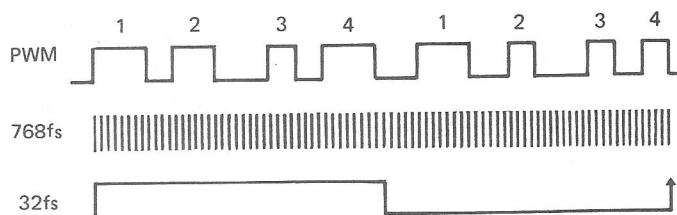
4. MASH Processing

Performs MASH processing on the L channel data from the first stage to determine 4 levels out of 11 levels and transfers them to the PWM block. Does the same processing for the R channel data.



5. PWM Processing

Pulse Width Modulates the 4 level data from the MASH block at a rate of 768fs, then separates the L and R channel data with 32fs clock. The L and R channel outputs of the PWM have the same phase, and are applied to the next stage at the rising edge of the 32fs clock.



10) Pin Description of MN6623A

● IC508: MN6623A (Digital Filter and D/A Converter)

Pin No.	Mark	I/O Division	Function
1	MLD	I	Command load input (load: L)
2	RSTB	I	Reset signal input (reset at "L")
3	IE	I	I ² S select terminal ("H"=I ² S)
4	TP1	—	TEST terminal (Not connected)
5	TP2	—	
6	TEST1	I	TEST terminal 1 (connected to GND)
7	TEST2	I	TEST terminal 2 (connected to GND)
8	NC	—	Not connected
9	NC	—	Not connected
10	AVDD4	I	Power supply (connected to +5V)
11	OUTR(-)	O	Rch data output, (-) terminal
12	AVSS4	I	GND terminal
13	AVSS3	I	GND terminal
14	OUTR(+)	O	Rch data output, (+) terminal
15	AVDD3	I	Power supply (connected to +5V)
16	NC	—	Not connected
17	AVDD2	I	Power supply (connected to +5V)
18	OUTL(-)	O	Lch data output, (-) terminal
19	AVSS2	I	GND terminal (analog system)
20	AVSS1	I	GND terminal (analog system)
21	OUTL(+)	O	Lch data output, (+) terminal

Pin No.	Mark	I/O Division	Function
22	AVDD1	I	Power supply (connected to +5V)
23	AVDD1	I	Power supply (connected to +5V)
24	DVSS1	I	GND terminal (digital system)
25	X2	O	Clock output
26	X1	I	Clock input
27	NC	—	Not connected
28	DVDD2	I	Power supply (connected to +5V)
29	DVSS2	I	GND terminal (digital system)
30	NSUB	I	Not used (connected to GND)
31	768fs	O	768 fs (Not used)
32	192fs	O	192 fs (Not used)
33	LRPOL	I	LR clock selector (Not used)
34	LR	I	L/R clock
35	SRCK	I	Serial data best clock
36	SRDATA	I	Serial data input (MSB first)
37	DVSS 3	I	GND terminal (digital system)
38	NC	—	Not connected
39	384 fs	O	384 fs (16.9344 MHz) output
40	PD	I	Power down terminal
41	MDATA	I	Command data input
42	MCLK	I	Command clock input

4. CIRCUIT OPERATIONS (TYPE C)

C4-1 AN8370S CIRCUIT OPERATION

The optical servo reads the signals recorded on a disk using the laser beam. AN8370S is the main IC of the optical servo system consisting of laser APC, RF signal amplifier, focus servo (including auto focus), tracking servo (including kick), drop output detector, and the control logic that controls the optical servo in this IC.

A block diagram of AN8370S is shown in Fig. 4-1.

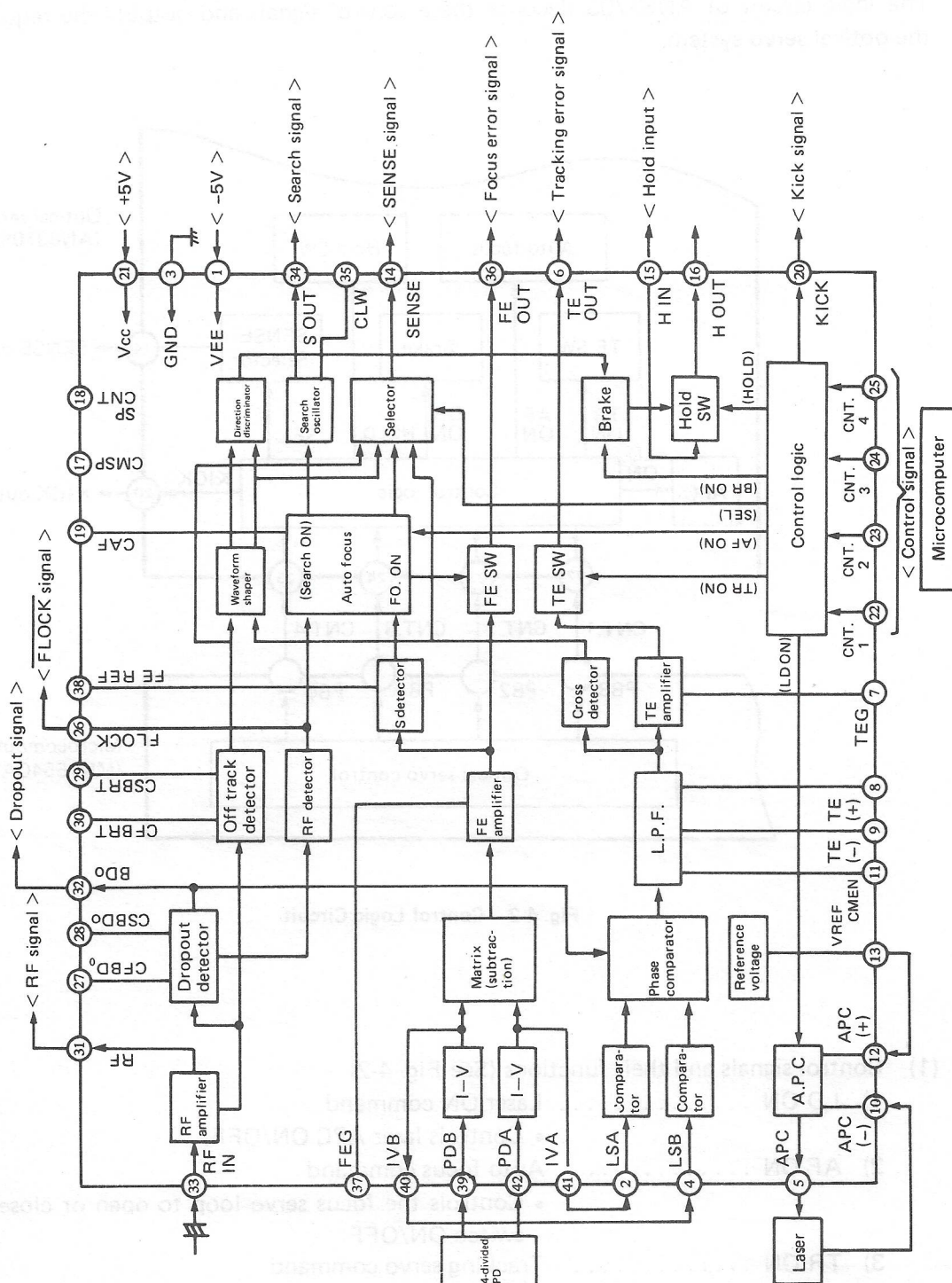


Fig. 4-1 AN8370S Block Diagram

C4-1-1 Control Logic Circuit

Control signals CNT1 to CNT4 are 4-bit parallel output signals from the microcomputer. The logic circuit of AN8370S decodes these control signals and outputs the required control signals to the optical servo system.

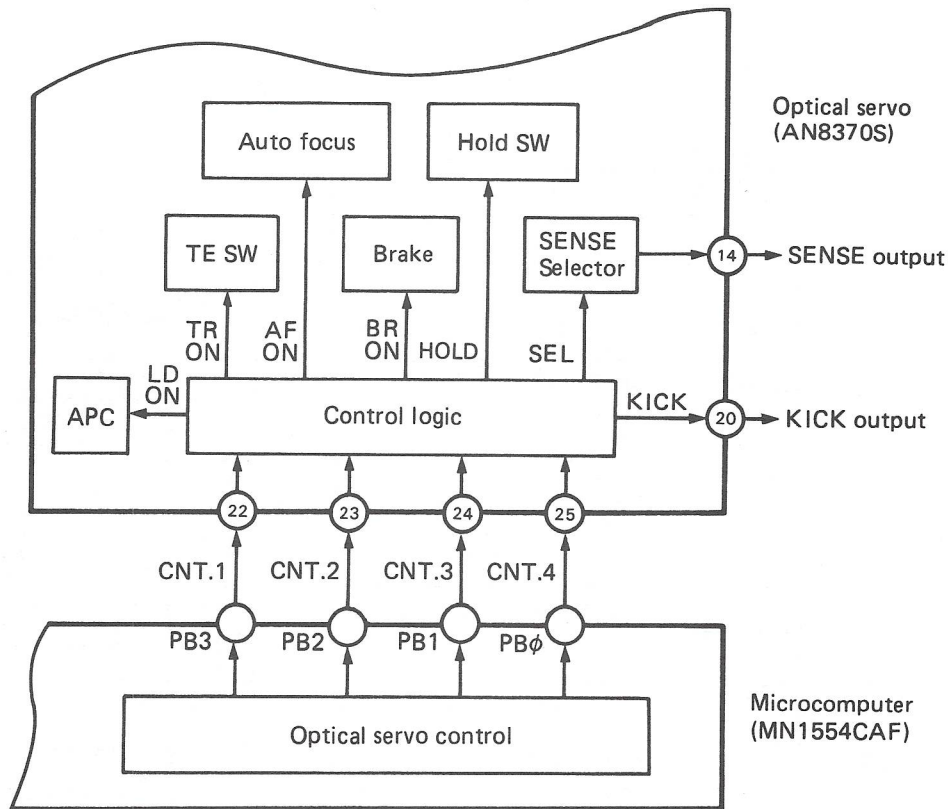


Fig. 4-2 Control Logic Circuit

(1) Control signals and their functions (See Fig. 4-2)

- 1) LD ON Laser ON command
 - Controls laser APC ON/OFF.
- 2) AF ON Auto focus command
 - Controls the focus servo loop to open or close, and the search OSC switch ON/OFF.
- 3) TR ON Tracking servo command
 - Controls the tracking servo loop switch.
- 4) HOLD Tracking hold command
 - Holds tracking error at preceding value during track kick.
- 5) BR ON Tracking brake command
 - After high-speed feed or kick, brakes the tracking actuator to absorb unnecessary vibration in a short time to facilitate tracking.

- 6) SEL SENSE signal select command
- Selects SENSE signal (that indicates optical servo status) to be sent from AN8370S to the microcomputer.
- SENSE signal changes depending on the system mode.
- 7) KICK Track kick command
- Controls track kick output signals.
- There are KICK FWD (outward) and KICK REV (inward), and three modes of 1 kick, 10 kicks, and 100 kicks according to kick distance.
- The logic circuit has $16 (2^4 = 16)$ decoding modes, of which eight are used as shown in Table 4-1.

(1 "H", 0 "L")

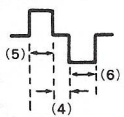
HEX No.	Mode	Logic input				Logic control signal output							Remarks
		CNT1	CNT2	CNT3	CNT4	LD ON	AF ON	TR ON	HOLD	BR ON	KICK	SENSE	
0	PLAY	0	0	0	0	ON	ON	ON	OFF	OFF	OFF	Off track	Play mode
3	BRAKE	0	0	1	1	ON	ON	ON	OFF	ON	OFF	Off track	Brake mode
4	KICK CROSS	0	1	0	0	ON	ON	OFF	ON	OFF	OFF	CROSS	
5	KICK REV	0	1	0	1	ON	ON	OFF	ON	OFF	ON (REV)	Shaping CROSS	
6	KICK FWD	0	1	1	0	ON	ON	OFF	ON	OFF	ON (FWD)	Shaping CROSS	
7	ACCESS	0	1	1	1	ON	ON	OFF	OFF	OFF	OFF	Shaping CROSS	Coarse feed
8	DISK DETECT	1	0	0	0	ON	ON	OFF	OFF	OFF	OFF	S latch	Disk detection
F	STANDBY	1	1	1	1	OFF	OFF	OFF	OFF	OFF	OFF	Shaping CROSS	Forced return

Table 4-1 Control Logic Table

LD = LASER DEVICE
 AF = AUTO FOCUS
 TR = TRACKING SERVO
 HOLD = TRACKING HOLD
 BR ~~ON~~ = TRACKING BRAKE
 SEL = SENSE

KICK = TRACK KICK

C4-1-2 Laser APC Circuit *Auto Laser power control output.*

Fig. 4-3 shows the laser APC circuit.

When a laser ON mode command (H, L, L, L) is received, pin 5 goes high to turn the laser on.

A loop circuit is formed by Q and D (the laser paired with a PIN diode) and the APC in the IC, and laser beam intensity is detected by the PIN diode in APC operation.

If laser beam intensity increases, the PIN diode's photo-electric current increases so that the pin to voltage rises (Pin 12 voltage is the reference voltage) to decrease laser beam intensity.

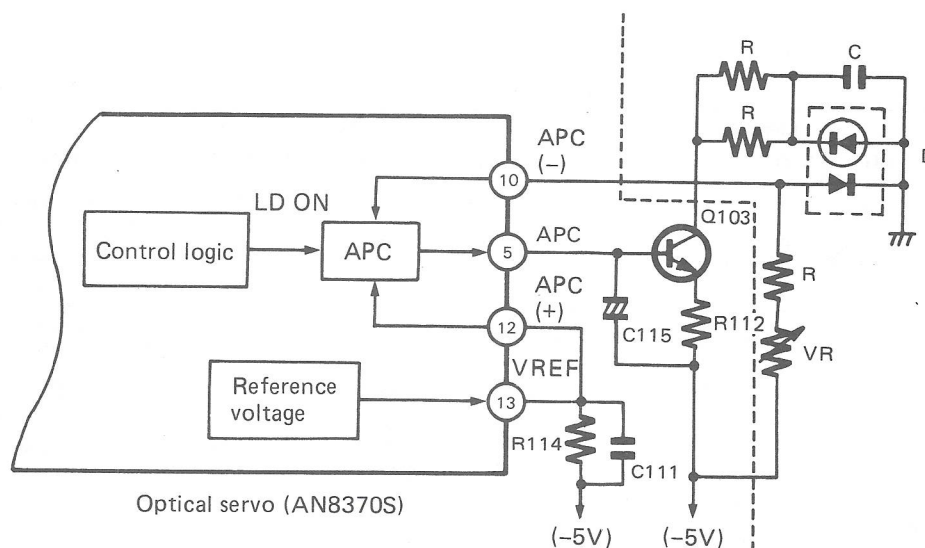


Fig. 4-3 Laser APC Circuit

C4-1-3 RF Amplifier and Dropout Detector Circuits

(1) RF amplifier circuit

RF signal is the total sum of all the 6-segment photo-detector ($A1 + A2 + A3 + A4 + B1 + B2$) of the optical pickup. The resulting RF signal is applied to pin 33, via Q, (Q is located in the optical pickup.) where it is amplified by an RF amplifier having gain of approx 34 dB.

The output of the RF amplifier appears at pin 31, and couples to pin 15 of IC AN8371S.

(2) Dropout detector circuit

The dropout detector detects signal dropouts in the RF signal, stabilizes the tracking servo against dropouts, and provides the defection signal to pin 32.

The detection dropout process is shown in Fig. 6-5. The level goes high if a dropout is detected.

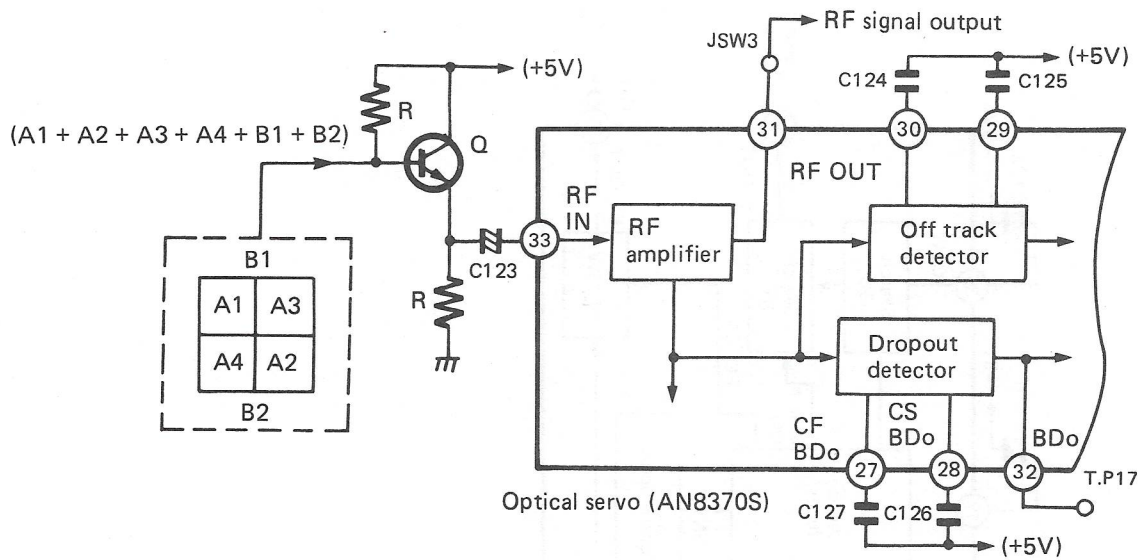


Fig. 4-4 RF Amplifier and Dropout Detection

(3) Off track detector

The off track detector detects the laser beam being off track.

When the tracking servo is turned off, the beam crosses the tracks during Search, Skip, mode even track-jump due to condition of disc surface, etc. In this case, the relationship between RF signal and off track detection signal is as shown in Fig. 4-6.

The off-track detector detects off-track using different time constants condenser connected to pin 29 (0.033 uf) and pin 30 (470 pf).

This detection signal is used as a control signal in the IC101 and also as SENSE signal for the micro-computer.

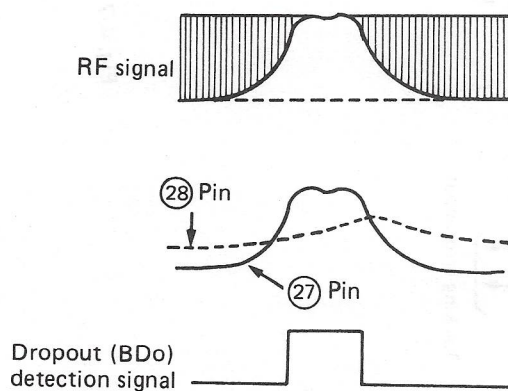


Fig. 4-5 Dropout Detection

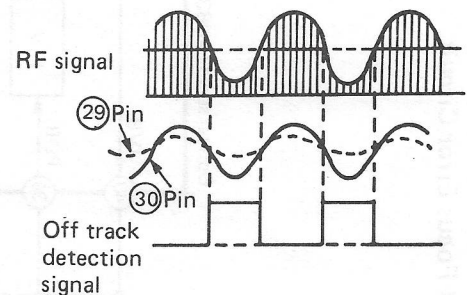


Fig. 4-6 Off Track Detection

C4-1-4 Auto Focus Circuit and Focus Error Circuit

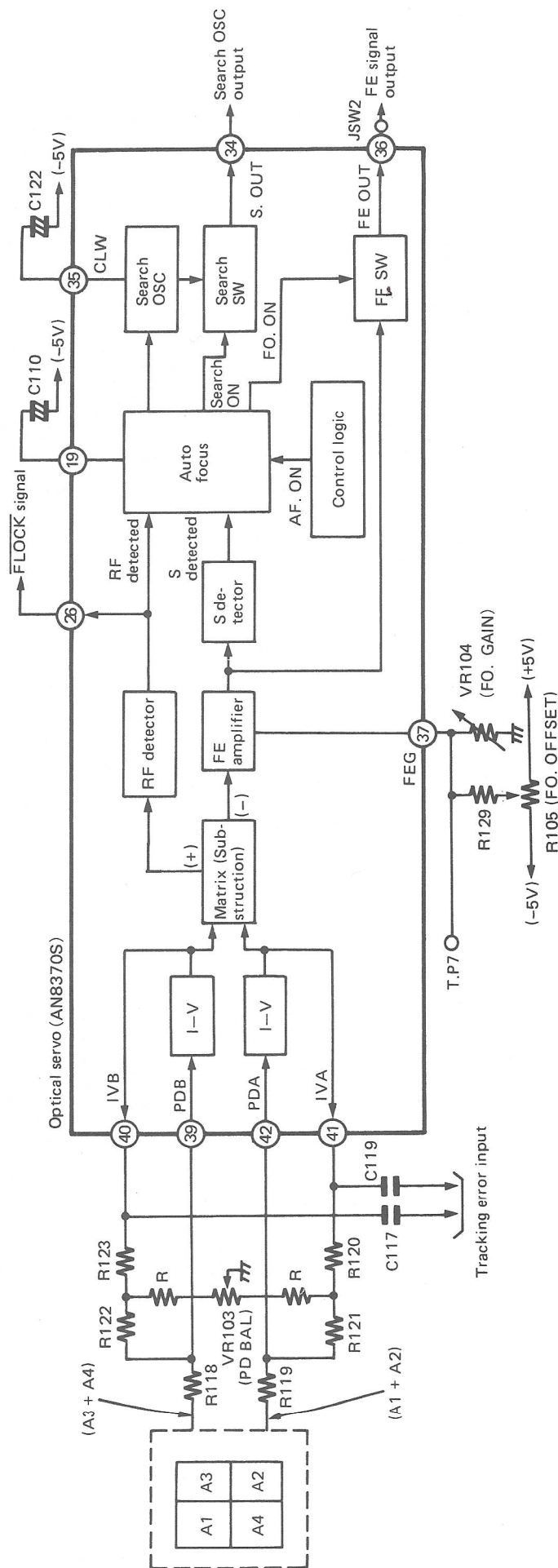


Fig. 4-7 Focus Error Circuit and Auto Focus Circuit

(1) Auto focus circuit

The auto focus is operated by the AF ON (auto focus command) signal output from the control logic circuit. It has two inputs, RF detection and S-curve detection, and regulates search SW and FE SW.

FE = FOCUS SIGNAL ERROR

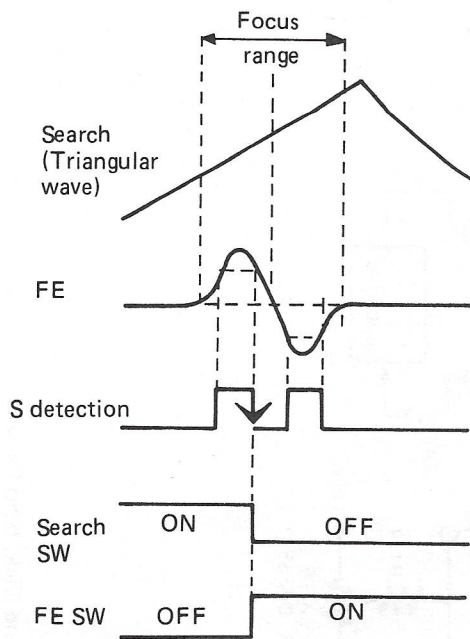


Fig. 4-8 Fo ON Timing

1) Disk detection

AF ON turns search SW on and FE SW off.

In this state, the search signal (triangular wave) drives the focus coil (raising or lowering the lens).

When the distance between the disk and the lens nears the focus length by focus search, the FE signal develops an S-curve, which is detected to turn search SW off and FE SW on to close the focus servo loop as shown in Fig. 4-8.

SW = SWITCH

Identifica la funzione
di "Search SW" = Ricerca
focus on e di loop
FE SW (FOCUS ERROR
CONTROL)

2) Re-focusing

The auto focus circuit checks the servo status by RF detection when the focus servo loop is closed. When RF detection is no longer possible due to defocusing, search SW is turned ON to start the focus search. The search signal is input to the microcomputer.

Whether the auto focus is working can be checked the voltage at pin 26 (F LOCK). (Low when the focus is correct.)

FIG. 4-7

(2) Focus error circuit

The diagonal sums ($A1 + A2$) and ($A3 + A4$) of the 4-segment photo-detector are input to pins 39 and 42 of IC101. The I/V converter converts the photo-detector output currents to voltages. An external resistor is used for gain setting. VR101 is a photo-detector balance (best eye) adjuster, for setting off the sensitivity difference of the photo-detector.

The difference between the two converted signals ($A1 + A2$) - ($A3 + A4$) is a focus error signal (FE). The FE signal applied to pin 36 (FE out) of IC101 via the FE amplifier and FE SW.

VR104 is the focus gain (FO GAIN) control, and VR105 the focus offset (FO OFFSET) control.

C4-1-5 Tracking Error Circuit and KICK, HOLD Circuits

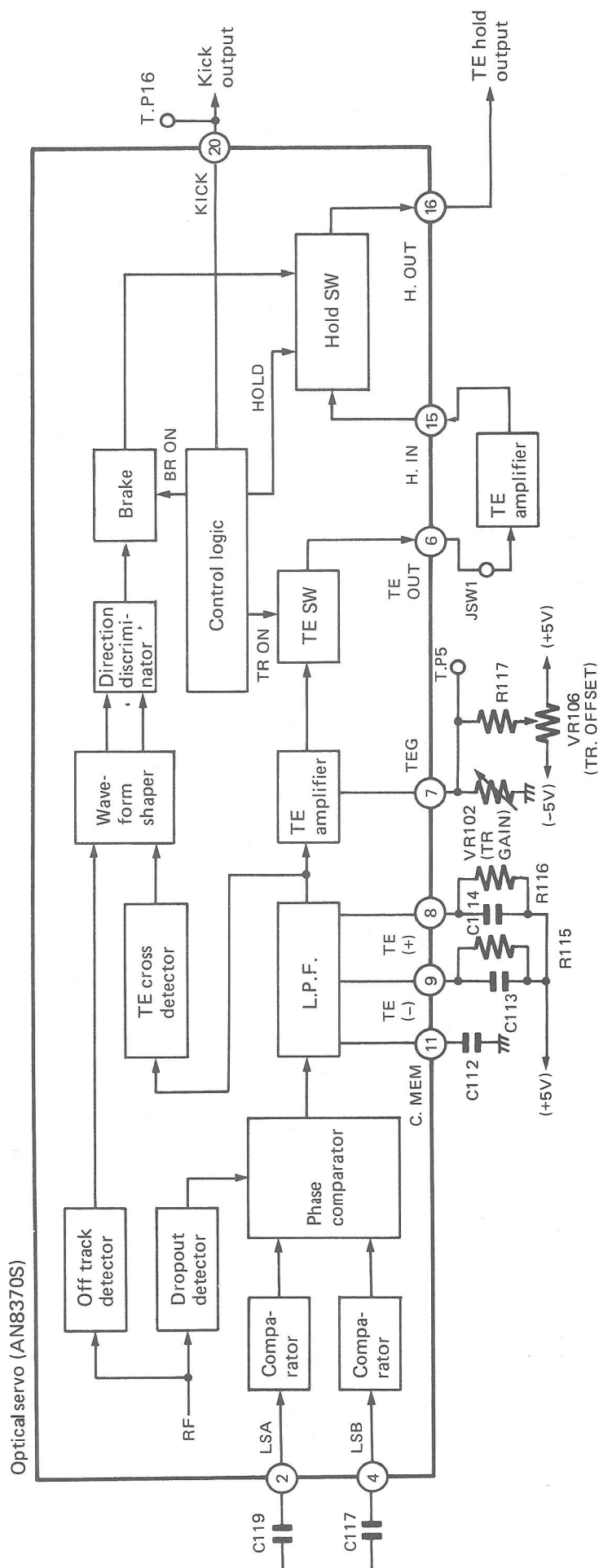


Fig. 4-9 Tracking Error Circuit and Kick, Hold Circuits

(1) Tracking error circuit

Tracking error circuit inputs the I-V converted signals to pin 2 and 4 via C119 and C117. The comparators shape the waveforms.

In the tracking error process of the single beam type, the phase difference between the two signals is detected and a pulse corresponding to the phase difference is generated as described in section 1-4. Deviating direction is judged from the polarity of the pulse. The generated pulse is smoothed by the LPF into an analog tracking error (TE) signal. This is how the phase comparator and LPF operate.

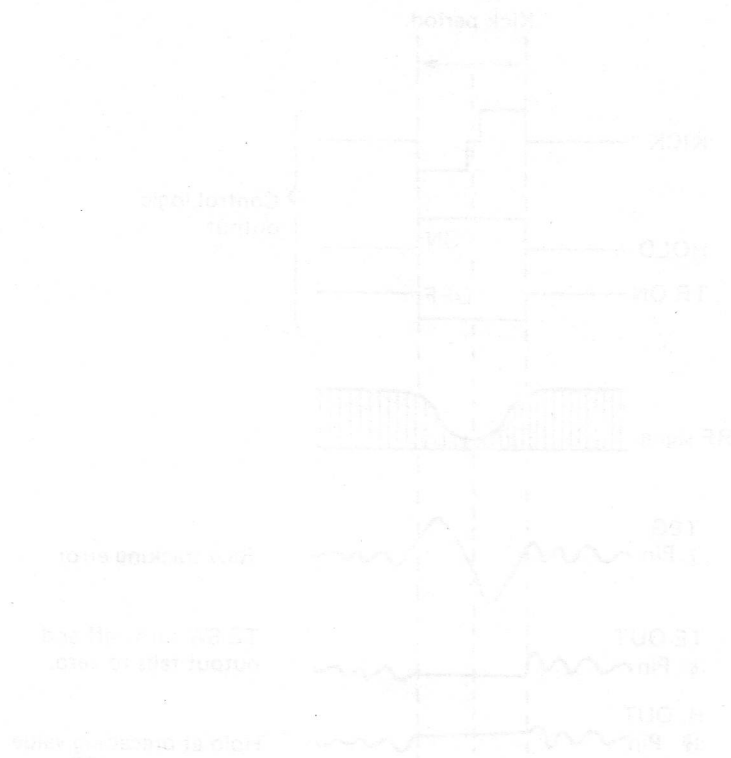
Then, the TE signal is output from pin 6 (TE OUT) via the TE amplifier and TE SW.

VR102 is the tracking gain (TR GAIN) control, and VR106 the tracking offset (TR OFFSET) control.

(2) Kick and hold circuits

Kick means jumping the beam from one track to another during a pause or manual search. The kick timing is set by a control signal from the microcomputer, and a kick signal is appeared pin 20. The waveform of each kick signal is shown in Fig. 4-10.

The hold circuit holds the preceding TE signal during kick made. TE SW is turned off during kick mode. Signal wave forms in pause kick are shown in Fig. 4-11.



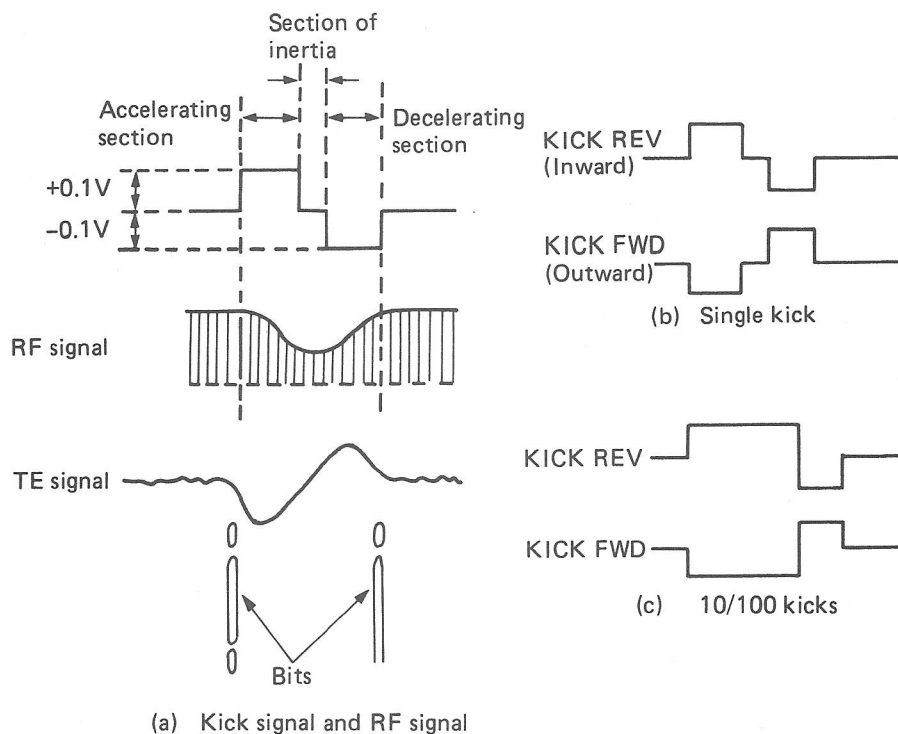


Fig. 4-10 Kick Signal Waveforms

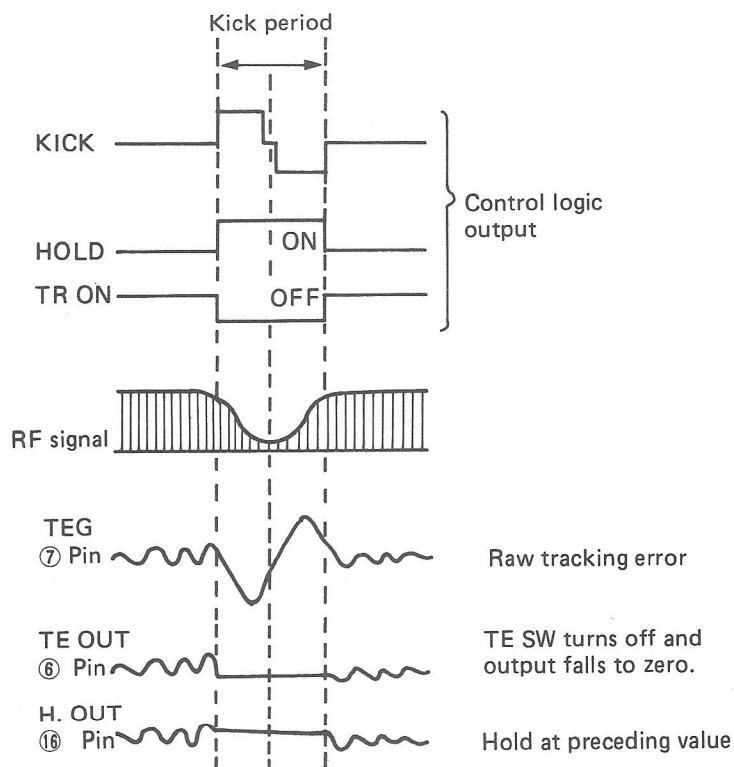


Fig. 4-11 Pause Kick

(3) Brake circuit

The brake circuit brakes the tracking actuator to quickly retract the tracking servo after rapid feed or kick (10/100). The process is shown in Fig. 4-12.

The brake circuit applies a single limiter (indicated by the dotted line in the figure) to the TE signal to absorb the energy after kick.

The figure shows the brake in the KICK REV mode. In the KICK FWD mode, the brake is applied with the single limiter inverted in polarity. Therefore, the brake circuit requires a kick direction discriminating signal, which is generated from a tracking cross signal and off track signal as shown in Fig. 4-13.

The cross signal phase leads in the KICK FWD mode and lags in the KICK REV mode. The brake direction is determined by detecting the above phase difference.

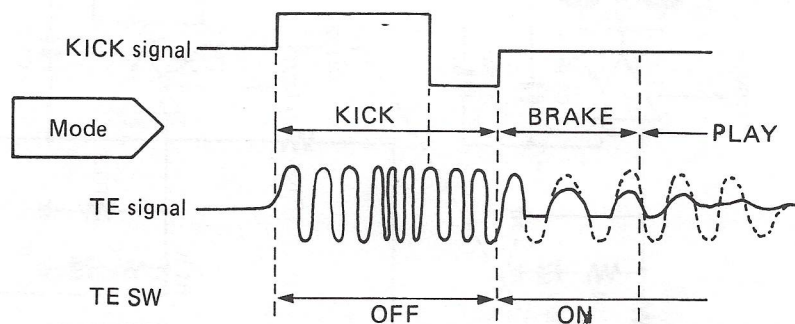


Fig. 4-12 Brake Timing

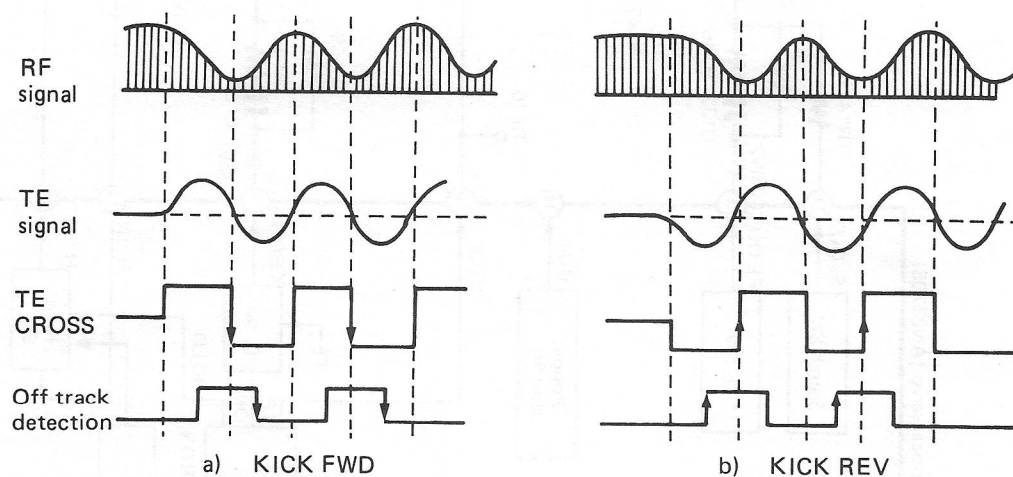


Fig. 4-13 Kick Direction Discrimination

C4-1-6 Focus Tracking Drive Circuit

Both servos have the same drive circuit except that the focus servo has a search circuit and that the tracking servo has a kick hold circuit.

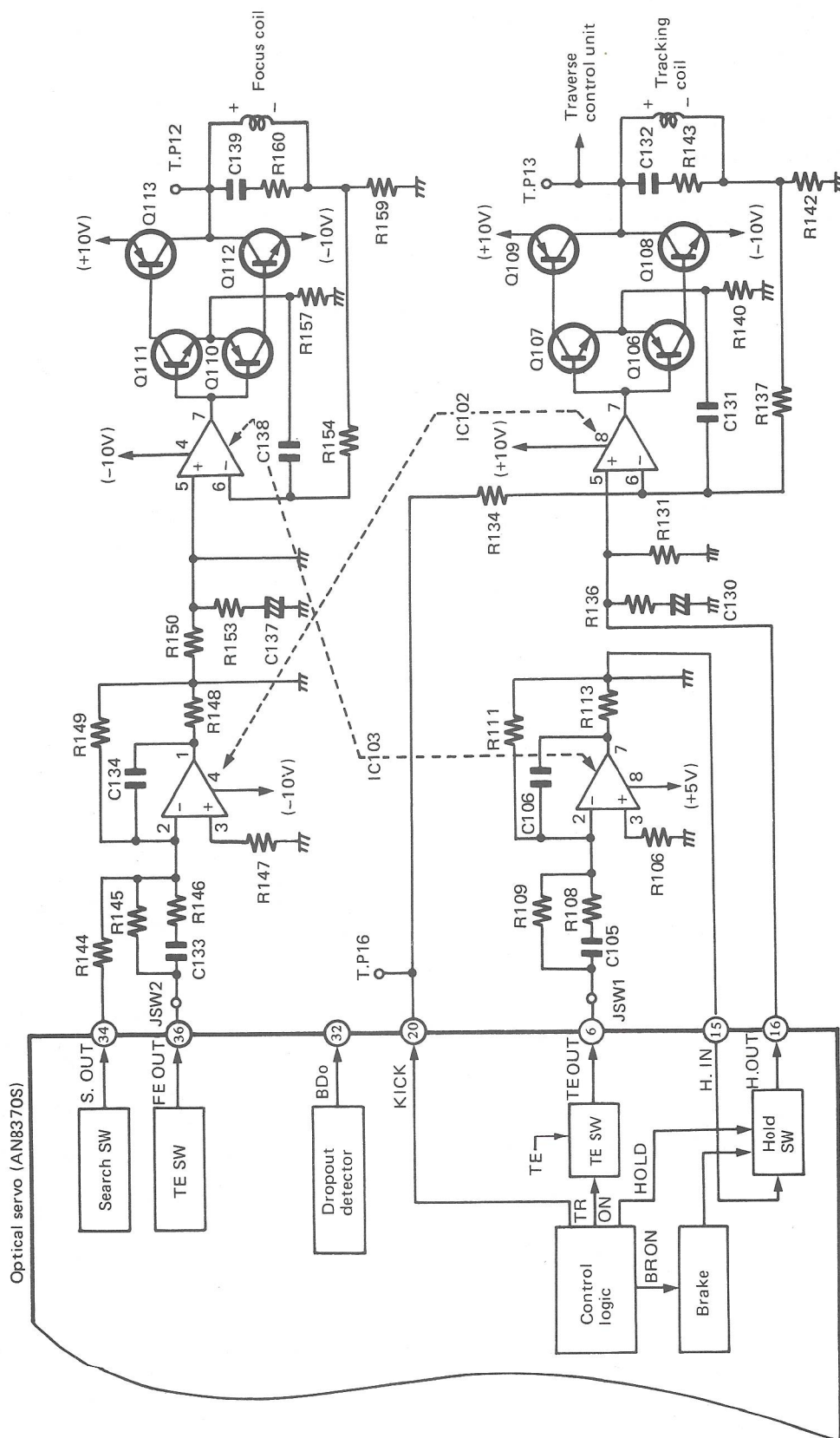
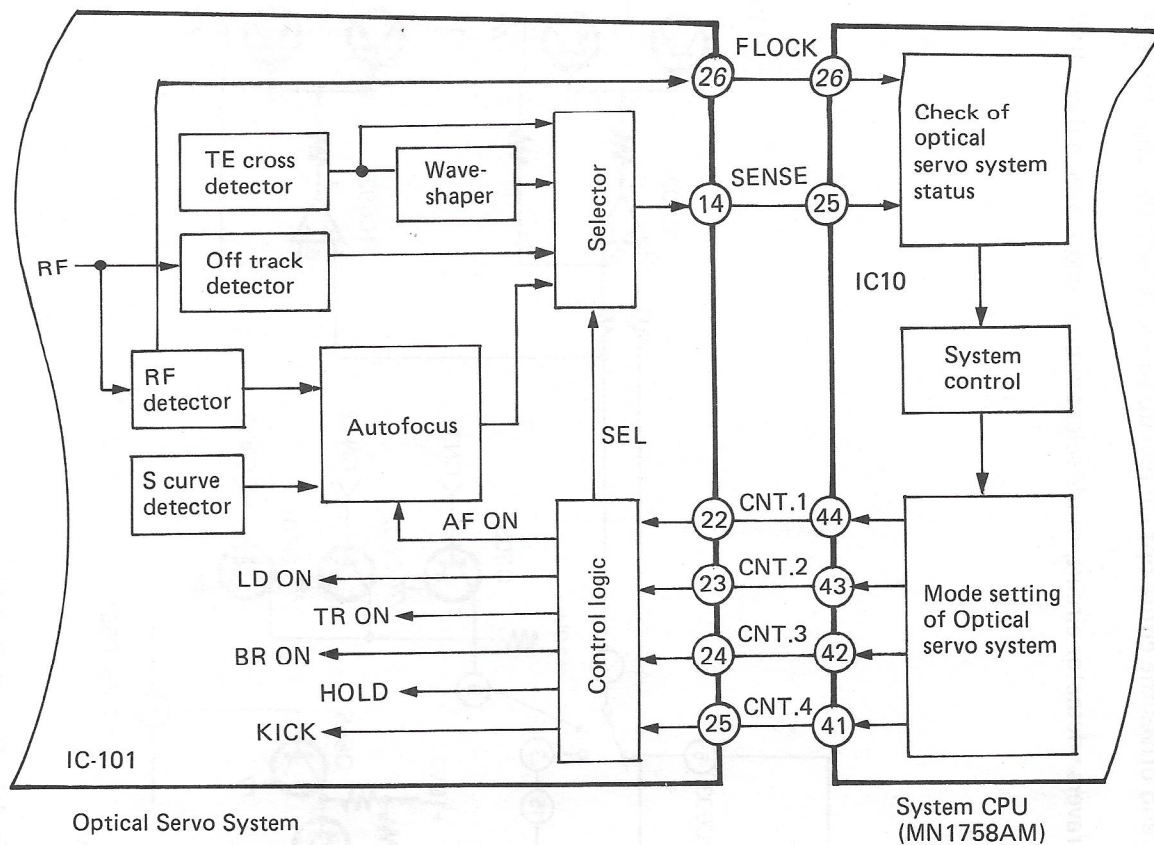


Fig. 4-14 Focus Tracking Drive Circuit

Relation with Servo IC and System CPU

The figure 4-15 shows the interconnection diagram of servo IC-101 (AN8370S) and system IC10. There are for buses for carrying signals from the system IC to AN8370S and two buses for carrying signal from AN8370S to the system IC.



P.49A Fig. 4-15 Connection Diagram of System CPU and Optical Servo System

- (a) Output signal from system: — Control logic signals (CNT1 — CNT4)
 (b) Input signal from system: — RF signal (FLOCK) Active "L"

Servo system status signal (SENSE) . . .

Type of SENSE signal:

- (1) Off-track signal
- (2) Tracking cross signal
- (3) Waveshaped tracking cross signal
- (4) Focus S-curve detection signal.

C4-2 TRAVERSE SERVO CIRCUIT OPERATION

The traverse servo detects the inclination of the objective lens and drives the entire optical pickup so that keeps the objective lens within the range of tracking servo.

The traverse servo circuit is shown in Fig. 4-1. It consists of a traverse detector circuit, mode select circuit, and traverse drive circuit.

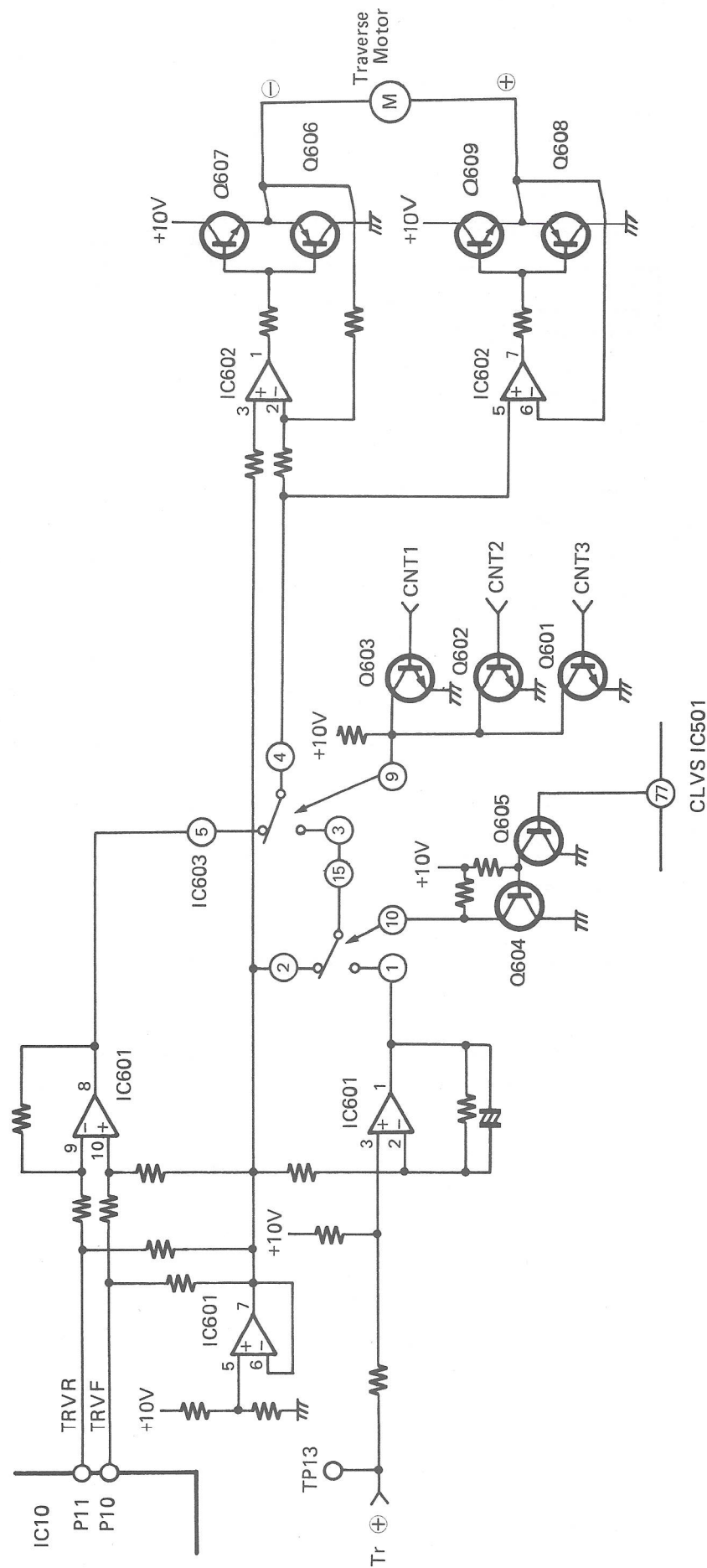


Fig. 4-1 Traverse Servo Circuit

C4-2-1 Traverse Detector Circuit

The tracking error signal is a signal of the positional deviation component of the disk as far as one turn of the disk is concerned. In a long period, however, DC voltage (servo error voltage) is generated as shown in Fig. 4-2. This DC voltage drives the traverse motor.

The traverse detector circuit in here range from the tracking coil output to the lowpass filter (pin 1 of IC601) output. Pin 8 of IC601 outputs forced drive signals, that is, forced forward (H) and reverse (L), by microcomputer commands.

Pin 7 of IC601 is set to a reference voltage of +5V.

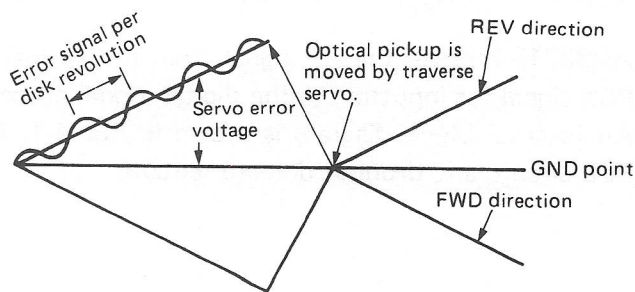


Fig. 4-2 Traverse Servo Voltage

C4-2-2 Mode Select Circuit

The multiplexer (IC603) is used for mode selection.

- 1) Servo error voltage selection (Play mode)

The output from pin 1, IC601 is applied to the drive circuit when pin 10 and pin 9 of IC603 are high.

- 2) Brake selection by reference voltage (KICK mode)

The output from pin 7, IC601 is applied to the drive circuit when pin 10 of IC603 is low and pin 9 of IC603 is high.

- 3) Forced forward/reverse selection (SKIP mode)

The output from pin 8, IC601 is applied to the drive circuit when pin 9 of IC603 is high.

The input to pin 9 of IC603 is controlled by CNT 1 to CNT3 of the microcomputer via the NOR circuit. Thus, pin 9 goes high when all of CNT1 to CNT3 are low (during disk play), or goes low when one of CNT1 to CNT3 is high (other than during disk play).

The input to pin 10 of IC603 is controlled by CLVS signal from pin 77 of LSI (IC501). Thus, pin 10 goes high when T11 is OK.

C4-2-3 Traverse Drive Circuit

The traverse drive circuit consists IC602, Q606 to Q609 to drive the traverse motor using the reference voltage applied to pin 3 of IC602 and the polarity corresponding to the voltage difference applied to pin 2. Q606 to Q609 comprise a dual direction drive amp. Q607 and Q609 are NPN and Q607 and Q608 are PNP. By controlling the polarity of the voltage applied to the Bases of ALL TRANSISTORS, it will determine whether the motor is supplied with a (-) or (+) voltage thus determining the direction of motor movement.

C4-3 AN8371S CIRCUIT OPERATION

AN8371S receives the RF signal from the optical servo IC (AN8370S), and generates the EFM signal and PCK signal for inputting to the digital signal processing LSI (MN6617).

An internal block diagram is shown in Fig. 4-1. AN8371S consists of an AGC circuit, data slice circuit, PLL circuit, and dropout detector circuit.

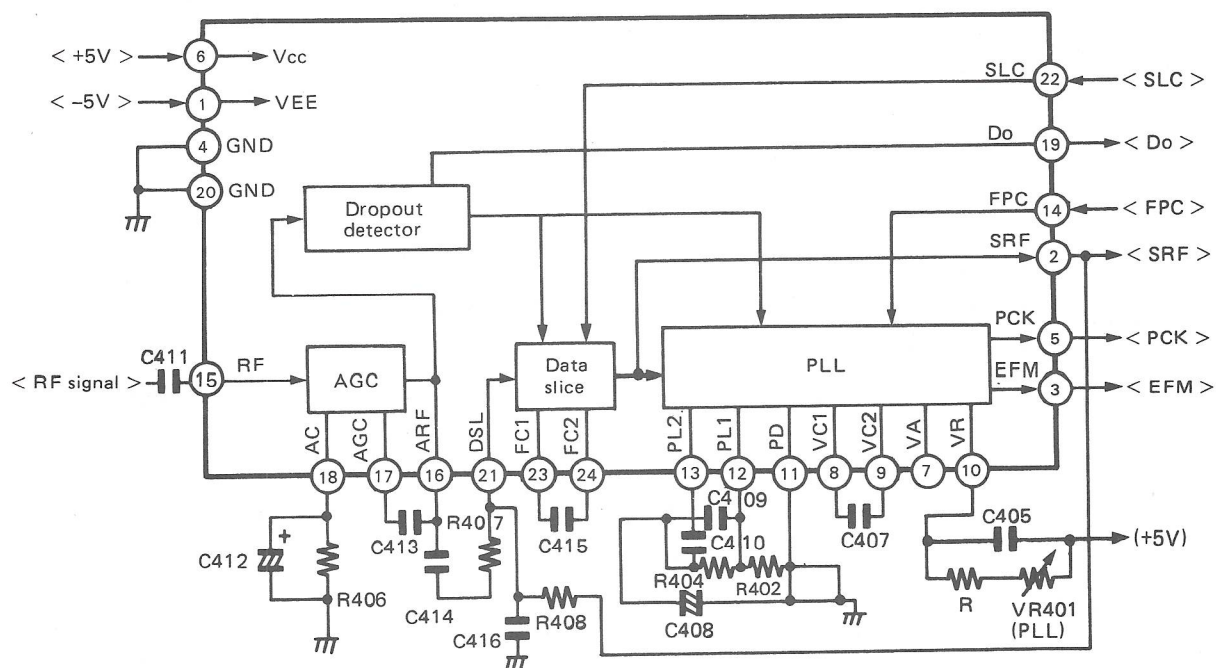


Fig. 4-1 AN8371S Block Diagram

C4-3-1 EFM Signal

The RF signal is an analog continuous signal as shown in Fig. 4-2.

Therefore, it cannot be processed as a digital signal unless it is shaped into a digital signal. This waveform shaping is performed by the data slice circuit, whose output signal is EFM signal.

C4-3-2 PCK Signal

The PCK signal is a clock signal synchronized with EFM signal.

The EFM signal includes a jitter component until it is written into the memory. Therefore, a clock signal synchronized with EFM signal is necessary in order for the digital signal processing circuit to accurately receive data.

The PCK signal is generated from the PLL circuit with an EFM signal input. Fig. 4-3 shows a timing chart of EFM signal and PCK signal.

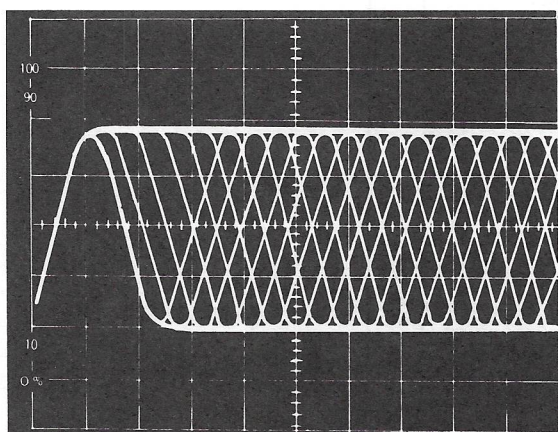


Fig. 4-2 RF Signal

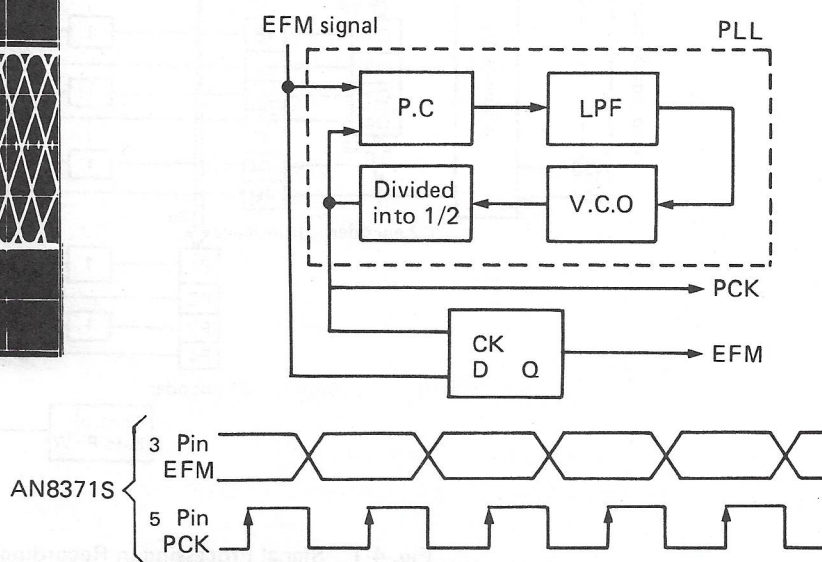


Fig. 4-3 EFM Signal and PCK Signal

C4-3-3 Other Signals

IC AN8371S outputs SRF signal and D0 signal besides the EFM and PCK signals. FPC and SLC signals are input signals to IC AN8371S.

The SRF signal is generated by data-slicing RF signal. The D0 signal is the dropout detecting signal in the RF signal (high at dropout). Both are input to the CLV control LSI (MN6617).

The FPC signal is the VCO control signal of the PLL circuit that is fed back from the LSI. The SLC signal is the slice level control signal from the microcomputer (MN1554CAC).

Their functions are described in Section 9-3.

C4-4 MN6617 CIRCUIT OPERATION

Prior to the circuit operation description, the signal processing of the recording system will be outlined.

C4-4-1 Signal Processing in Recording System

(1) Encoder

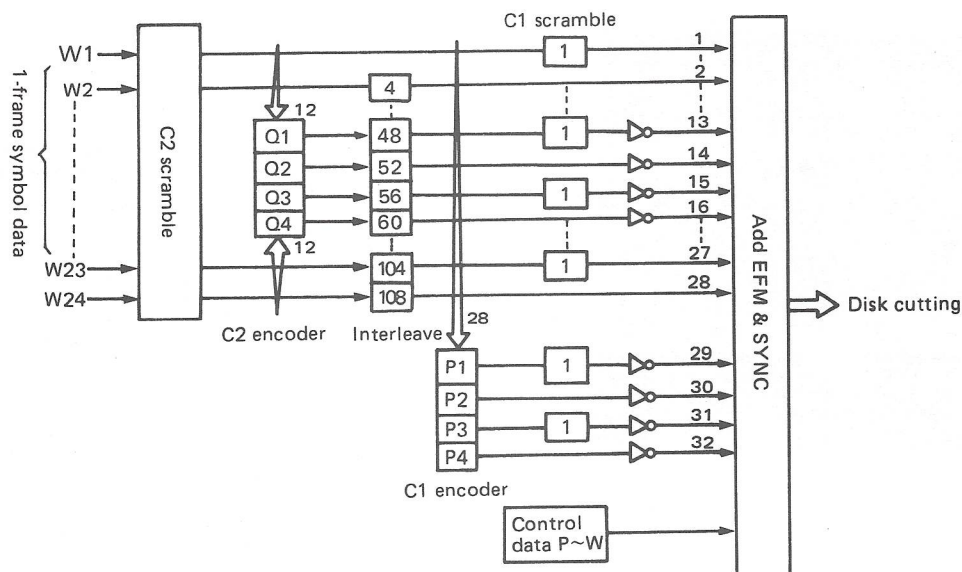


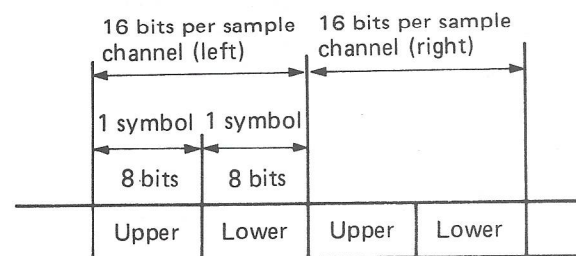
Fig. 4-1 Signal Processing in Recording System

Fig. 4-1 shows a block diagram of CD recording system signal processing.

A brief description of CD recording system signal processing is given in the following.

- 1) The CD sampling frequency is 44.1 kHz, and quantizing involves A—D conversion to 16 bits per sample.

The 16 bits are divided into upper 8 bits and lower 8 bits.



* 16 bits per sample are called a data word, and 8 bits per symbol are called symbol data. In other words, two symbols constitutes one data word.

Fig. 4-2 Symbol Data and Data Word